

MyoWare 2.0 Muscle Sensor

DEV-18977

Product Overview

10/11/2022

For the most up-to-date information, visit www.mouser.com or the supplier's website.

Description

SparkFun MyoWare 2.0 Muscle Sensor is an Arduino-compatible, all-in-one Electromyography (EMG) sensor from Advancer Technologies. This sensor is redesigned from the ground up with a new, easy-to-use, compact design. The MyoWare 2.0 muscle sensor is upgraded with the latest and greatest chipset, improving sensor performance and reliability.



The MyoWare 2.0 muscle sensor measures muscle activity through the electric potential of the muscle, commonly referred to as surface Electromyography (EMG or sEMG for short).

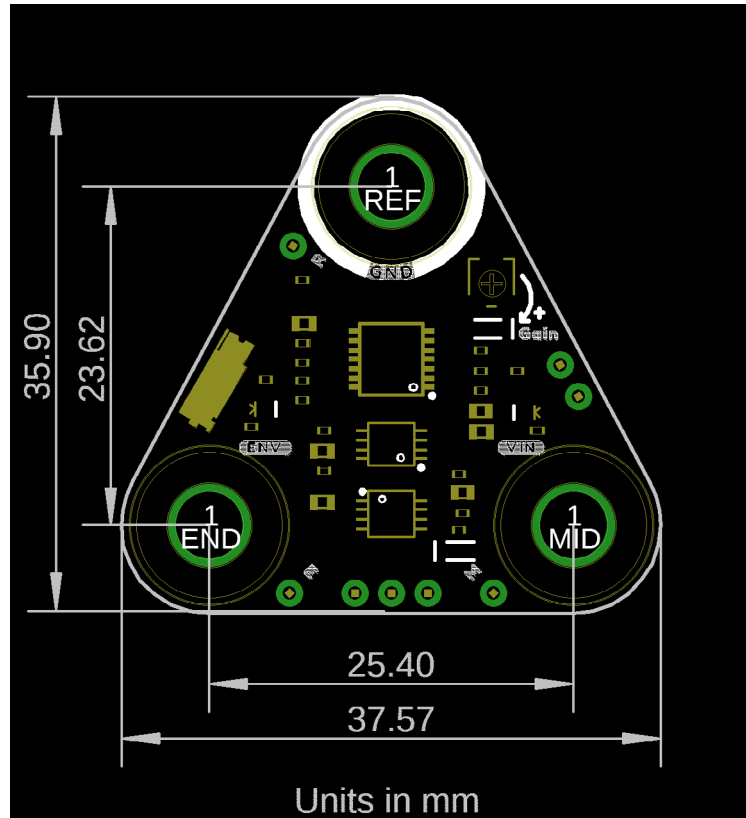
When the brain signals the muscle to flex, it sends an electrical signal to the muscle to start recruiting the motor. The harder the flex, the more motor units are recruited to generate great muscle force. The greater the number of motor units, the more electrical activity of the muscle. This muscle sensor will analyze a muscle's filtered and rectified electrical activity and output a signal (0V-VIN volts, where VIN signifies the power source voltage) that represents how hard the muscle is being flexed.

The MyoWare 2.0 muscle features a 3.3V to 5V single-supply voltage range, three output modes, reverse polarity-protected power pins, and indicator LEDs. This sensor also features reverse polarity protection, adjustable gain, and 250pA, maximum 1nA input bias current.

Features

- Wearable design
- Supply voltage:
 - 2.27V minimum voltage
 - 3.3V to 5V typical voltage
 - 5.47V maximum voltage
- Input bias current:
 - 250pA, maximum 1nA
- Reverse polarity protection
- Three output modes:
 - Raw EMG
 - Rectified
 - Envelope
- Expandable via shields
- MyoWare 2.0 muscle sensor form factor:
 - 3x female snap pins (power and EMG envelope output)
 - 3x male snap pins (input electrodes)
- LED Indicators:
 - VIN
 - ENV
- Reference electrode jumper
- Specially designed for microcontrollers
- Adjustable gain
- 37.57mm x 35.90mm board dimensions

Board Dimensions



Application Overview



Mouser Part Number

[View Part](#)

To learn more, visit <https://www.mouser.com/new/sparkfun/sparkfun-myoware-2-muscle-sensor/>